

Self-Check Sheet

Table of contents

Week 1: Kickstart with R	1
Week 2: Vectors and Matrices	2
Week 3: Data Frames and Lists	3
Week 4: Operators, Conditions, and Functions	3
Week 5: ggplot2 (part 1)	4
Week6: Revisiting the Essentials	4
Week7: Wrangling Data with dplyr and tidyr	5
Week 8: ggplot2 (part 2) & Custom Functions	5
Week 9: Reproducible Analysis with Quarto	6

Notes:

- Elements appear in the chronological order we saw them in class.
- Elements marked with * were mentioned in class but are **not required** to be mastered.

Week 1: Kickstart with R

```
sqrt()  
ls()  
library()  
mean()  
q()  
?  
getwd()  
file.exists()  
readr::read_csv()  
head()  
View()  
readr::write_csv()
```

dim()
typeof()
sd()
summary()
*hist()
*plot()
log2()

*text()
*abline()

Week 2: Vectors and Matrices

help()
NA
typeof() as.numeric()

c() :
seq()
paste0()
rep()
is.vector()
length()
names()
str()
tail()
is.na()
sum()
median()

matrix()
rbind()
cbind()
nrow()
ncol()
rownames()
colnames()
is.matrix()
rowSums()
colSums()
rowMeans()
colMeans()

```
*set.seed()  
round()  
*rnorm()  
shapiro.test()  
*curve()  
*dnorm()
```

Week 3: Data Frames and Lists

```
data.frame()  
is.data.frame()  
unlist()  
NULL  
factor()  
merge()  
  
list()  
names()  
  
read.table()  
unique()  
wilcox.test()  
*boxplot()
```

Week 4: Operators, Conditions, and Functions

```
<-  
=  
[]  
$  
[[]]  
+, -, *, /, ^  
<, <=, >, >=, ==, !=  
!, &, |  
%in%  
any()  
all()  
which()  
if()  
if ( ) {} else {}  
ifelse()  
print()
```

```
abs()  
table()
```

Week 5: ggplot2 (part 1)

```
install.packages()  
remove.packages()  
  
ggplot()  
aes()  
geom_point()  
labs()  
geom_smooth()  
geom_boxplot()  
geom_violin()  
geom_histogram()  
geom_bar()  
theme_grey()  
theme_bw()  
theme_classic()  
theme_bw()  
theme()  
ggsave()  
  
as.data.frame()  
theme_minimal()  
*element_text()
```

Week6: Revisiting the Essentials

```
read.csv()  
write.table()  
write.csv()  
readxl::read_xlsx()  
readxl::read_xls()  
writexl::write_xlsx()  
saveRDS()  
readRDS()  
save()  
  
t.test()
```

Week7: Wrangling Data with dplyr and tidyr

```
|>
*%>%

tibble()
as_tibble()

pivot_longer()
pivot_wider()
drop_na()

select()
starts_with()
ends_with()
contains()
matches()
mutate()
rename()
filter()
arrange()
desc()
count()
summarise()
group_by()
ungroup()

*pull()
*if_else()
*slice()
*slice_head()
*slice_tail()

*readr::read_csv2()
*extract()
*GGally::ggpairs()

cor()
```

Week 8: ggplot2 (part 2) & Custom Functions

```
theme_set()
theme_update()
scale_color_manual()
scale_fill_manual()
```

```

scale_shape_manual()
scale_x_continuous()
scale_y_continuous()
xlim()
ylim()
geom_hline()
geom_vline()
facet_wrap()
facet_grid()
geom_label()
*ggrepel::geom_label_repel()
*ggpubr::stat_compare_means()

theme_minimal()
*everything()
*sub()
*rank()
*position_jitter()
paste()

*ggpubr::stat_pvalue_manual()
geom_col()

*patchwork::wrap_plots()
*patchwork::plot_annotation()

```

Week 9: Reproducible Analysis with Quarto

```

# H1
## H2
### H3
**Bold**
*Italic*
`code`
`r nrow(iris)`
Superscript1
Subscript2
![] (path/to/image.png)
- unordered list
  - unordered sub-list
1. ordered list

```

```
left_join()  
theme_light()  
  
aov()  
*TukeyHSD()  
  
*bartlett.test()  
*kruskal.test()
```